Micro Services Architecture

- Typical System Design
- Microservice Architecture
    - API gateway
    - Services
    - Management or orchestration
    - Interservice communication
        - Synchronous
        - Message-oriented middleware (Async)
    - Data management
    - Observability

🚪 API Gateway

Single entry point that routes client requests to the appropriate microservices.

Responsible For **:**

- Routes requests
- Handles authentication
- Applies rate limiting
- Logs requests

Management or orchestration

Manages deployment, scaling, and availability of services across infrastructure.

Responsible For**:**

- Deploy services across servers/nodes
- Restart failed services
- Auto-scale based on traffic

Interservice Communication:

1. Synchronous
2. Asynchronous

🗄️ Data Management

Each service manages its own data and database independently.

Database per service

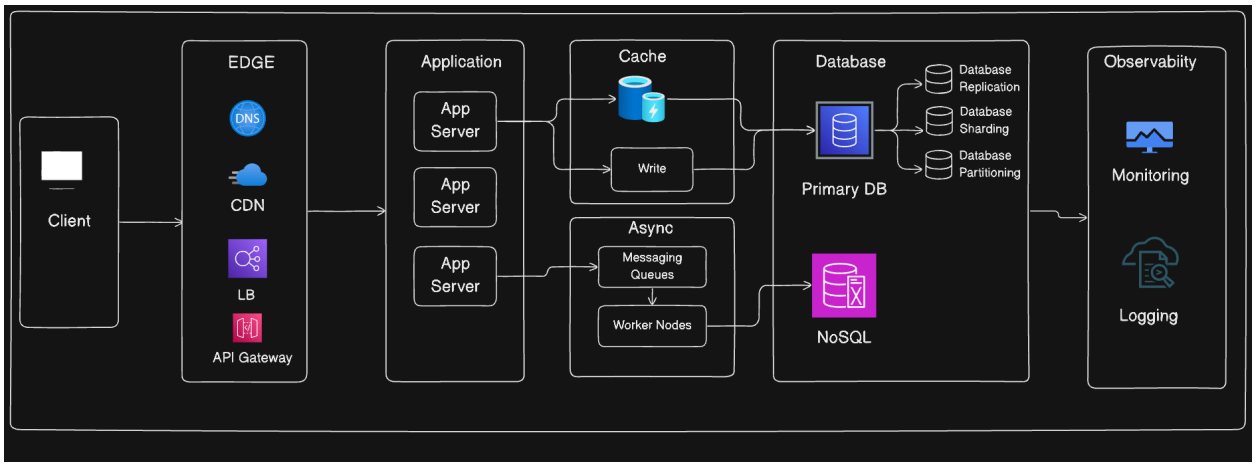
Different DB types possible

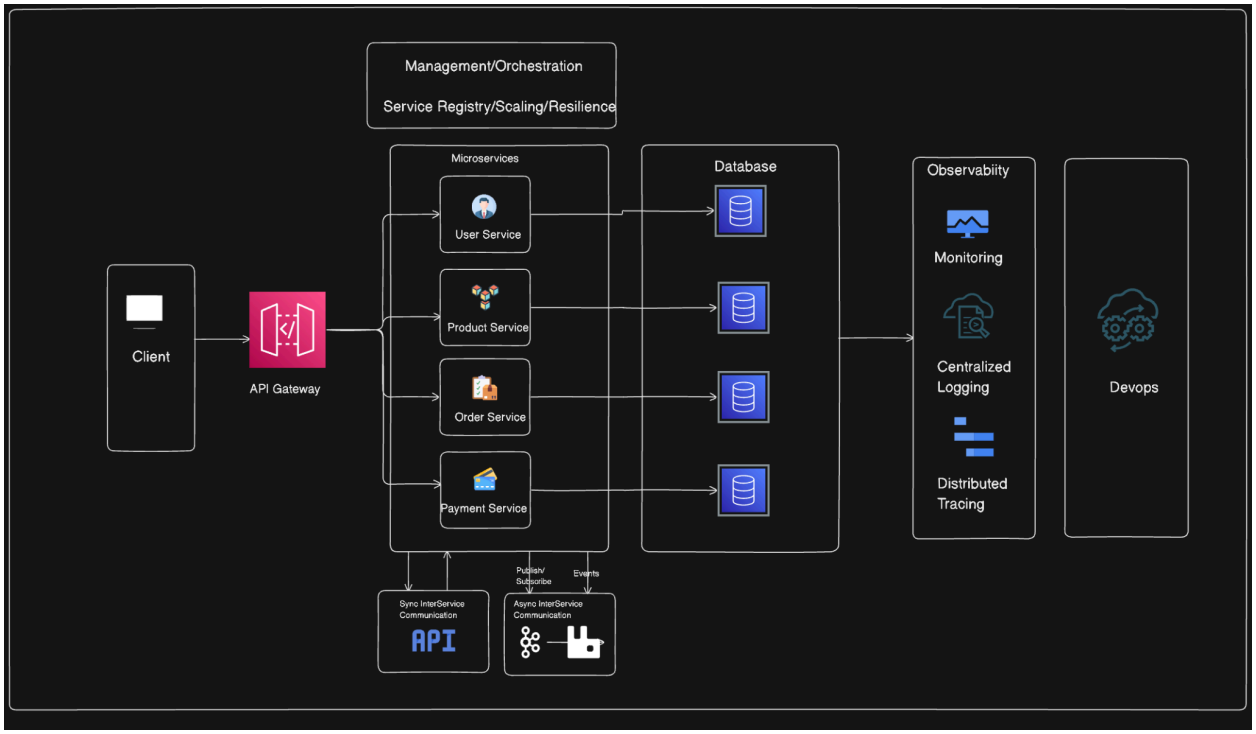
No shared database

📊 Observability

Provides visibility into system behavior through logging, monitoring, and tracing.

- Centralized logging
- Performance monitoring
- Request tracing across services




Management/Orchestration
Service Registry/Scaling/Resilience
Microservices
User Service
Product Service
Order Service
Payment Service
Database
Observabiity
Monitoring
Centralized Logging
Distributed Tracing
Devops
Client
API Gateway
Publish/ Subscribe
Events
Sync InterService Communication
API
Asyno InterService Communication